

The background of the slide features a dark blue map of Asia. Overlaid on the map are various financial data elements, including green and red bar charts, line graphs, and numerical values such as '112.63', '58.43', '00.47', '75.46', '47.13', '226.43', '2.15', '1.22', '3.01', '112.63', '58.43', '200.2', '4,263', '4,123', '14,025', '7,859', '53,135', '56.14', '10.56%', '20%', '3.46%', '11.59%', '18.15%', '112.63', '14.025', '7.859', '3.46%', '20%', '11.59%', '18.15%', '112.63', '14.025', '7.859', '3.46%', '20%', '11.59%', '18.15%'.

What is the outlook for the six biggest Asian Economies amid the energy crisis?

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COMP 2501

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Executive Summary

1. Data
2. Data Science Questions
3. Observing countries' indicators evolution and the revised commodity prices forecasts from the IMF
4. Regression Analysis: Measuring the impact of Supply Chain disruptions across countries
5. Limitations and Opportunities for further Research



Why this topic is interesting and important

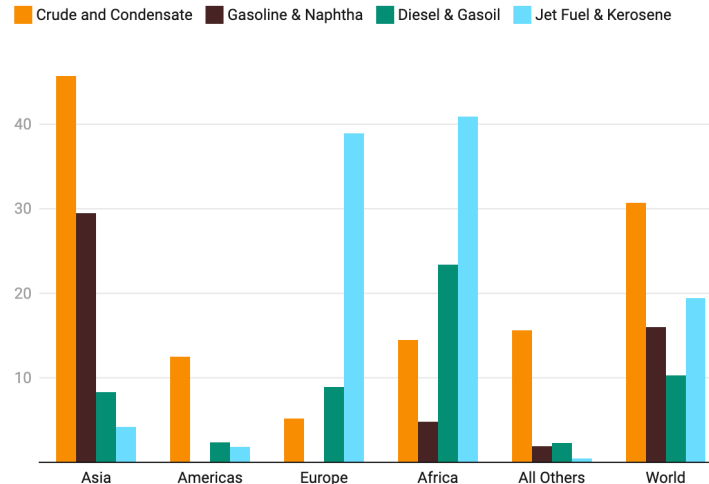
As an Economics student, it's an opportunity to learn more about the continent and how its countries shape the World Economy.

Asia is the future: it represents 60% of the world's population, and 50% World GDP.

Asia is one of the regions that is most affected by the crisis in the Middle East: almost 50% of the continent's oil imports pass through the currently blocked strait of Hormuz.

Strait Oil Dependency varies by Products and Continent

Strait of Hormuz import as a % of global import, 2025



Source: Asia Global Institute, March 2026

In that context, it's important to study the "fundamentals" to determine the resilience of countries.

Trump and Xi's meeting in Beijing on May 14-15 is one of the most awaited geopolitical events of the year.



Picture: ALAMY. Trump arrives in Beijing to meet President Xi, May 2026.

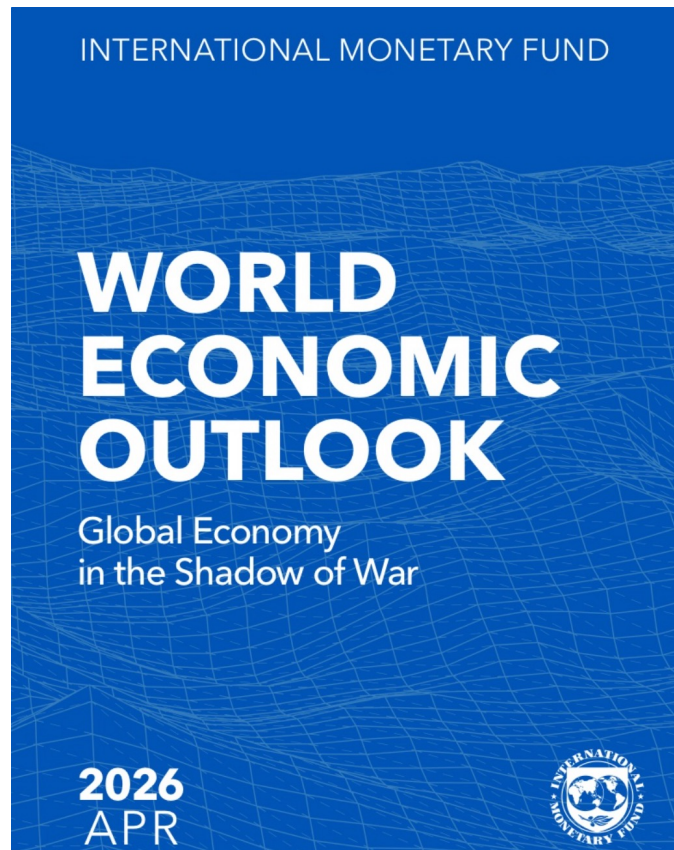
The outcome of the discussions will have an impact on all major global issues, including AI, climate change, etc.



The Data

The IMF's bi-annual report and dataset:

A range of indicators (GDP Growth, Inflation, Government Debt, etc.) on more than 190 countries, spanning the years 1980 to 2026, with forecasts up to 2031.



A	B	C	D	AD	AE	AF	AG	AH	AI	AJ	AK	
DATASET	SERIES_C	COUNTRY	COUNTRY	1980	1981	1982	1983	1984	1985	1986	1987	
IMF.RES:VHKG.NGDIHKG			Hong Kong	578.112	631.256	649.477	688.542	757.412	762.827	847.53	961.061	1
IMF.RES:VHKG.NGDIHKG			Hong Kong	10.34	9.193	2.886	6.015	10.002	0.715	11.104	13.396	
IMF.RES:VHKG.NGDIHKG			Hong Kong	143.62	173.577	196.005	217.282	261.992	278.127	320.526	394.771	
IMF.RES:VHKG.NGDIHKG			Hong Kong	28.862	31.055	32.291	29.907	33.511	35.7	41.076	50.623	
IMF.RES:VHKG.PPPCHKG			Hong Kong	33.435	39.963	43.657	48.095	54.815	56.953	64.551	75.012	
IMF.RES:VHKG.NGDIHKG			Hong Kong	24.843	27.497	30.179	31.557	34.59	36.46	37.819	41.077	
IMF.RES:VHKG.NGDIHKG			Hong Kong	114251.4	121864.1	123474.7	128699.4	139463.4	138685.7	152277.3	171150.4	1
IMF.RES:VHKG.NGDIHKG			Hong Kong	18595.94	19835.01	20097.16	20947.56	22699.54	22572.96	24785.18	27857.03	2
IMF.RES:VHKG.NGDIHKG			Hong Kong	28383.39	33509.07	37263.3	40613.45	48240.99	50564.87	57589.52	70302.74	8
IMF.RES:VHKG.NGDIHKG			Hong Kong	5703.963	5995.182	6139.013	5590.161	6170.503	6490.372	7380.118	9015.195	1
IMF.RES:VHKG.PPPCHKG			Hong Kong	6607.692	7714.794	8299.781	8989.741	10093.22	10354.32	11597.97	13358.44	1
IMF.RES:VHKG.NGAFHKG			Hong Kong									
IMF.RES:VHKG.PPPCHKG			Hong Kong	0.253	0.27	0.276	0.285	0.3	0.292	0.314	0.343	
IMF.RES:VHKG.PPPECHKG			Hong Kong	4.296	4.343	4.49	4.518	4.78	4.883	4.965	5.263	
IMF.RES:VHKG.NIDIHKG			Hong Kong	34.853	34.94	30.983	26.669	24.418	21.5	23.432	26.068	
IMF.RES:VHKG.NGSIHKG			Hong Kong	33.871	32.835	30.824	27.615	31.642	30.669	32.509	36.452	
IMF.RES:VHKG.PCPIHKG			Hong Kong	19.555	21.408	23.767	26.133	28.358	29.367	30.417	32.142	
IMF.RES:VHKG.PCPIHKG			Hong Kong	4.444	9.479	11.016	9.958	8.514	3.556	3.575	5.671	
IMF.RES:VHKG.PCPIHKG			Hong Kong		22.4	24.7	27.4	28.8	29.7	30.9	33.2	
IMF.RES:VHKG.PCPIHKG			Hong Kong			10.268	10.931	5.109	3.125	4.04	7.443	
IMF.RES:VHKG.TM.FHKG			Hong Kong	18.483	12.853	-1.486	10.085	14.666	6.776	13.395	28.534	
April 2026 WEO	Countries	Country Groups	Commodity Prices	Country Group Composition								

- The second dataset is of the commodity prices



Data Science Questions

Data Visualization

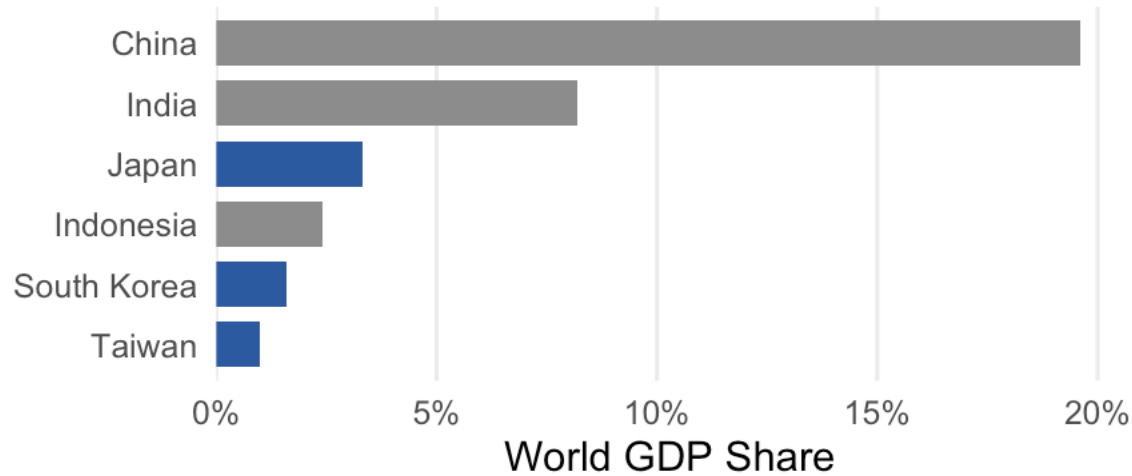
- What are some of the characteristics of the Asian economies under study as reflected in their growth trajectories and economic indicators?
- How has the IMF revised its commodity prices forecasts with the war?

Causal Analysis

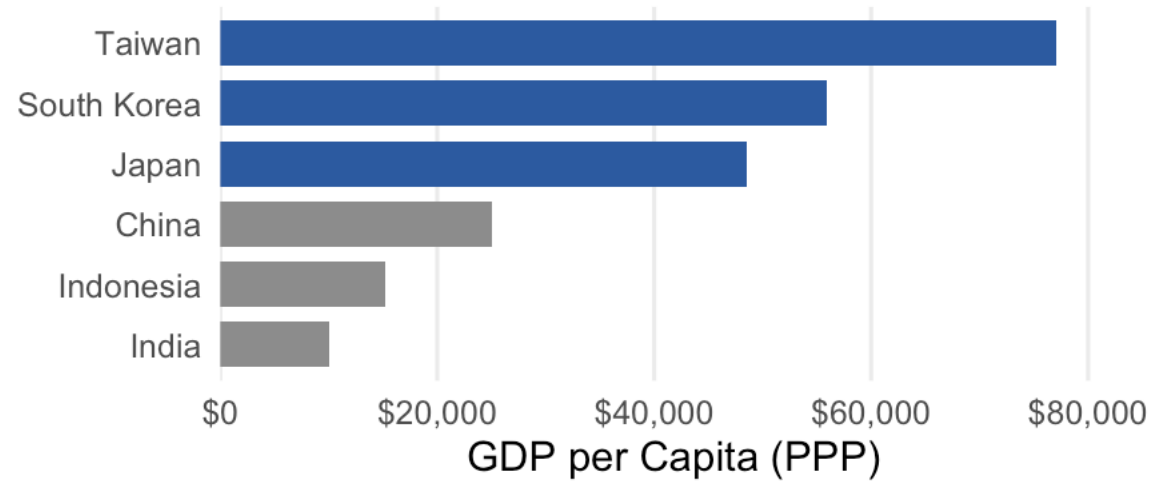
- Which country will likely be the most impacted by the crisis? Does the impact differ across emerging and advanced economies?



The biggest and richest economies in 2025



■ Advanced ■ Emerging



■ Advanced ■ Emerging

```
WEO_Asia_tidy |>
  filter(Year == 2025 & COUNTRY %in% selected_countries) |>
  summarise(
    Total_World_GDP_Share = sum(world_share_gdp)
  ) |> pull(Total_World_GDP_Share)
```

1 36.154

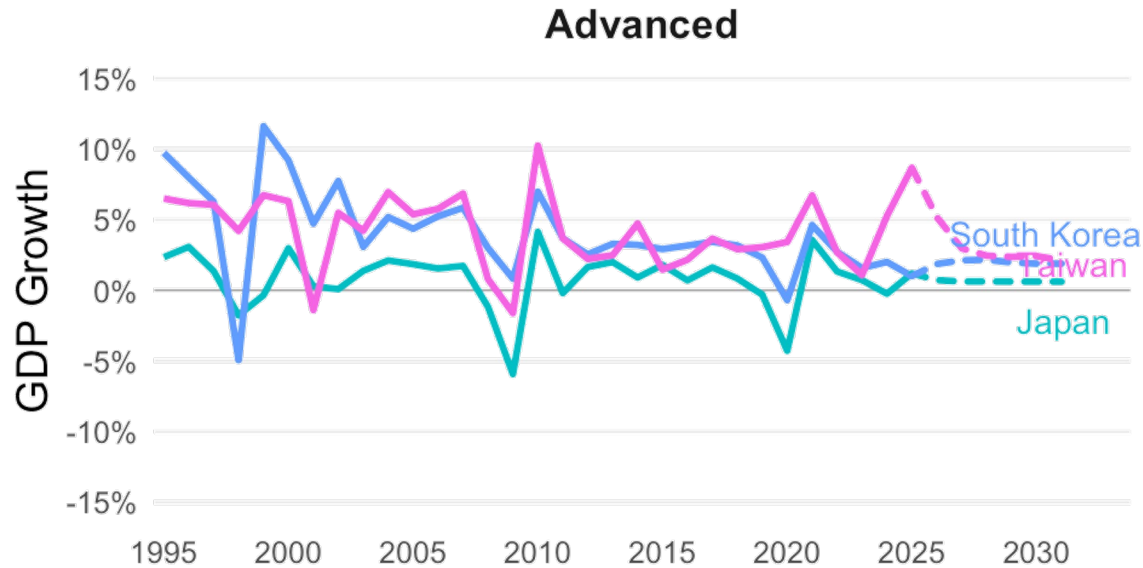


Asian Economies by Category and Characteristics

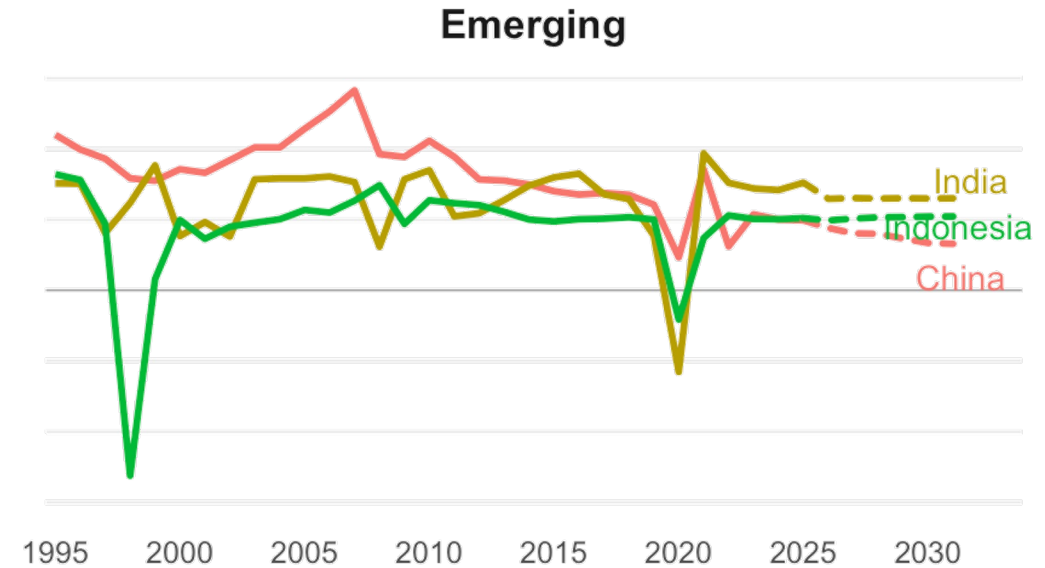
Advanced (6)		Emerging (19)	
Japan	- +200% debt as percent of GDP, financed largely nationally - Stuck with low growth and an ageing population	China	- Population of 1.4bn, 2nd largest economy in the world: leading many industries sectors, 50% of the World's manufacturing - Faces challenges of economic slowdown
South Korea	- Export-oriented economy (6th largest exporter) with the powerful Chaebol - Demographic crisis	India	- Population of 1.4bn - Manufacturing powerhouse, but becoming increasingly service oriented (50% of jobs in the 2020s)
Taiwan	- Leader in electronics exports since the 1970s - Known for its semi-conductor industry: TSMC is one of the most important companies in the world, sharing the stage with NVIDIA	Indonesia	- Devastated by the 1997 crisis from which it took decades to recover because of the stringent conditions imposed by the IMF - Commodity abundant country (nickel, cobalt, LNG), yet those resources are used domestically rather than exported
Geopolitically, the Advanced economies are U.S. allies while the Emerging ones are members of BRICS+ (even though India and Indonesia follow a strategy of multi-alignment). But Trump's trade policy has reshuffled the deck.			



Growth



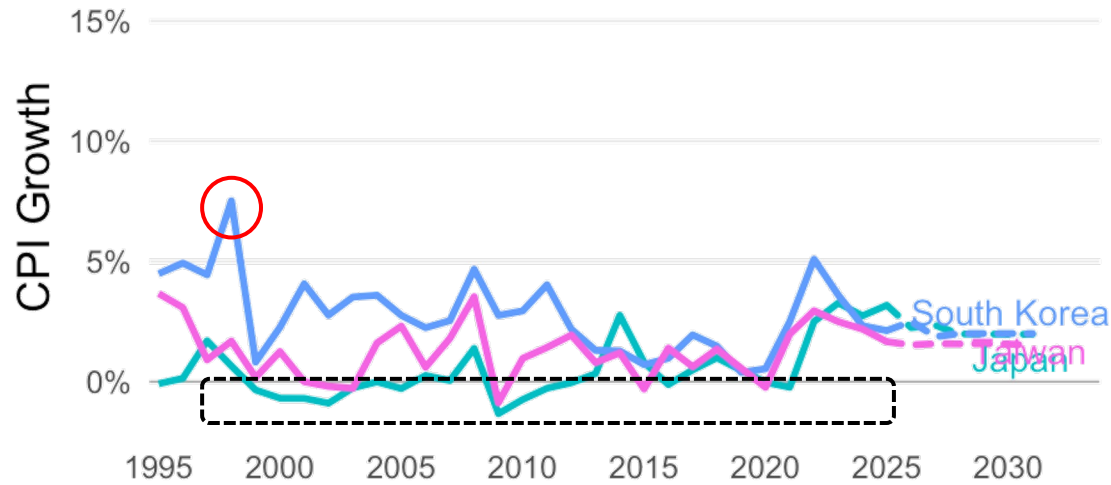
- Japan has the lowest growth of all, dangerously close to 0%.
- Taiwan's growth boosted by the AI bubble and is expected to come back to normal.
- South Korea was highly impacted by the Asian Financial Crisis of 1997 and quickly picked up after.



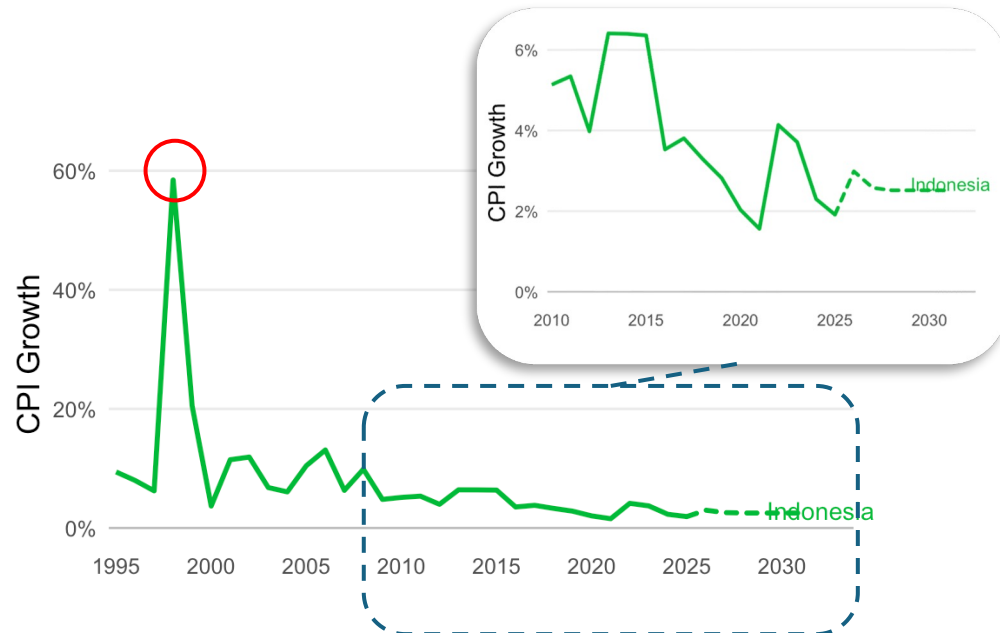
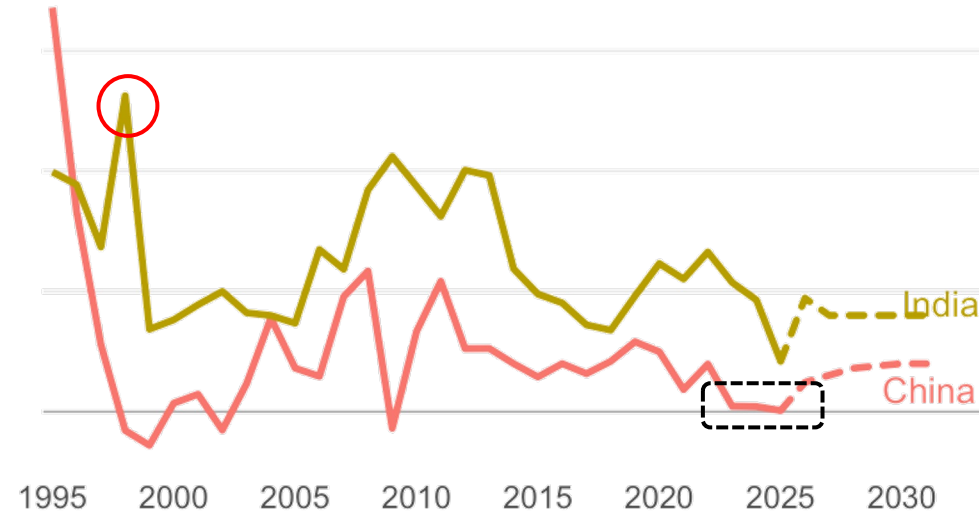
- Indonesia was much more sensitive to the shock of 1997, while it was India that suffered most economically from the Pandemic. China showed resilience throughout with a continuous two-digit growth.
- Today, though China is showing more weakness with growth coming down, outpaced by India and Indonesia.

Inflation

Advanced



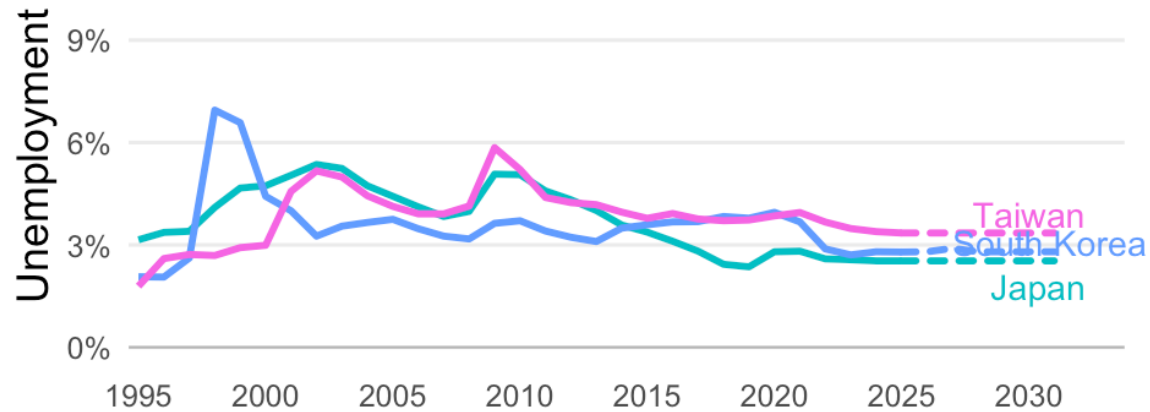
Emerging



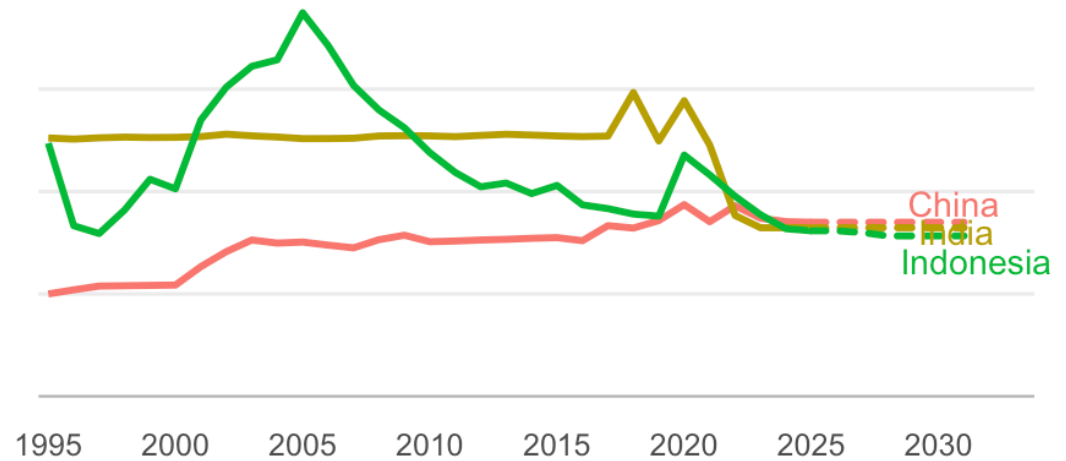
- We can identify which countries have suffered most from higher prices during the 1997 crisis: Indonesia, India, and South Korea. ○
- Inflation expectations adjusted upwards for 2026 with the Iran crisis, more importantly in Emerging Economies than in Advanced ones.
- Despite the common view, inflation is important for growth – as long as it's contained – which explains why Japan's has even tended to be negative and showing that China's economy has slowed down recently. □

Unemployment

Advanced



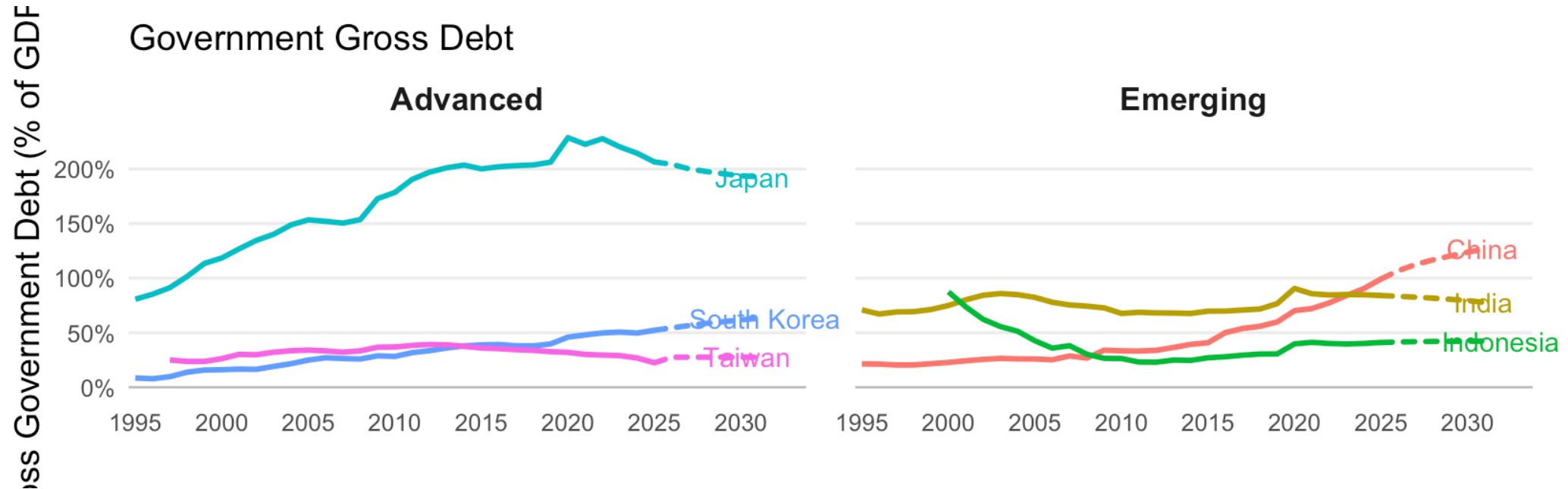
Emerging



- Despite its lower growth, Japan also has the lowest unemployment rate.
- Again, it can be seen how South Korea suffered from Financial the crisis but quickly recovered.

- India and Indonesia's economies outpacing China is reflected in this indicator too, with their economies experiencing a surge since the Covid Pandemic.
- Meanwhile it seems that China and India's unemployment rates have followed stable paths since 1995, while Indonesia suffered and recovered from the 1997 crisis.

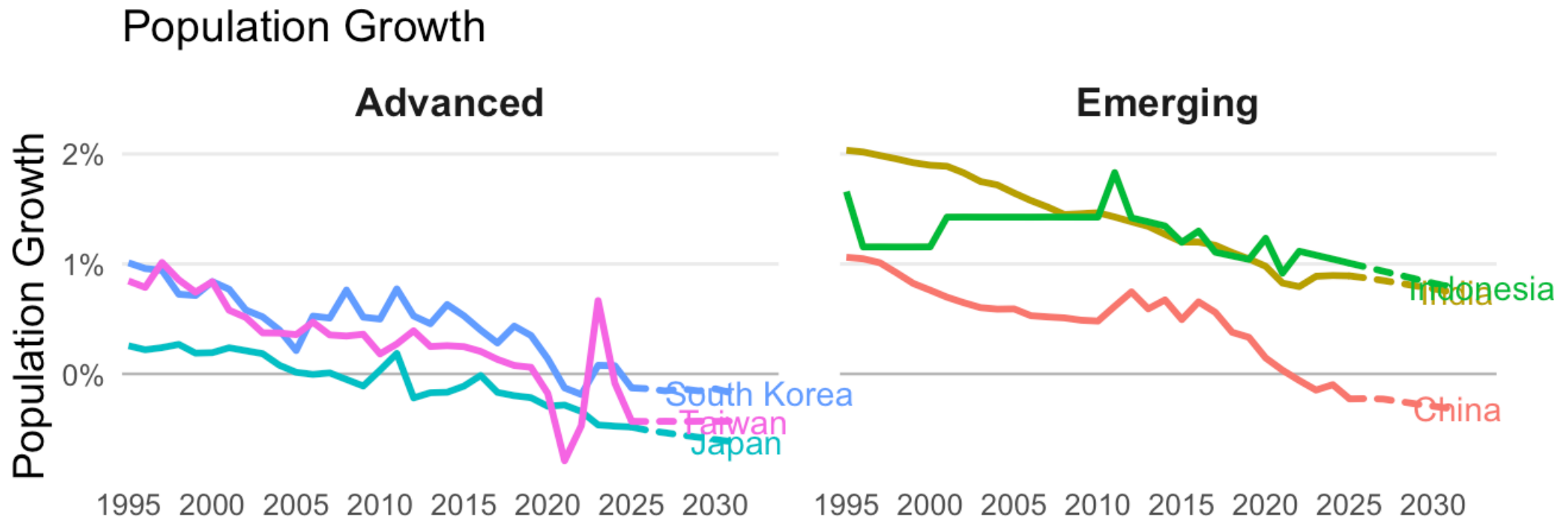
Debt as a Percentage of GDP



- Again, China's Debt since to be heading in the wrong direction at a fast pace, compared to its BRICS+ peers.
- South Korea's Gross debt is also projected to increase.

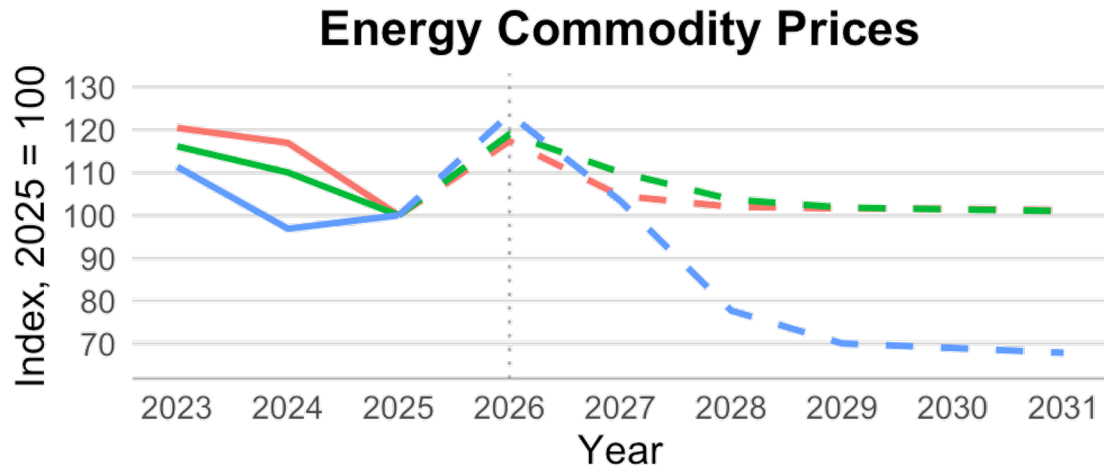


Population Growth



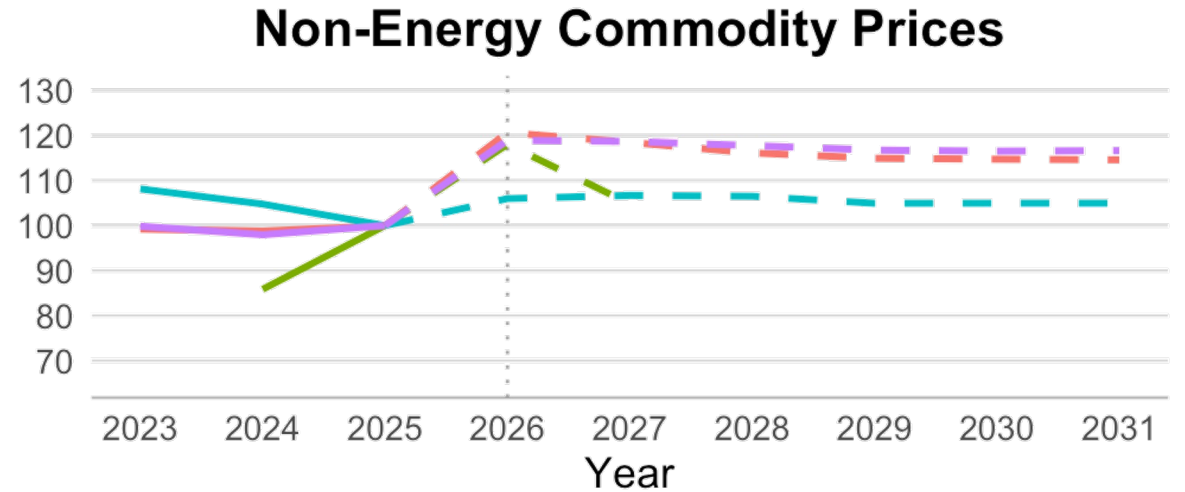
- Advanced economies all suffer from declining populations, as is the case for China, while India and Indonesia are still growing.

Global Supply chains disruptions: Forecasts from the IMF and the World Bank



— Brent Crude USD/b — Energy, Commodity price index — LNG Asia

- Prices are projected to highly increase in 2026 and gradually return back down to 2025 level.
- By 2028, LNG prices are even expected to be lower than in 2025.



— All Commodities Index* — Fertilizers Index — Food Index — Metal Index

- Prices seem to be more sticky across all categories

Shortfalls of the dataset:

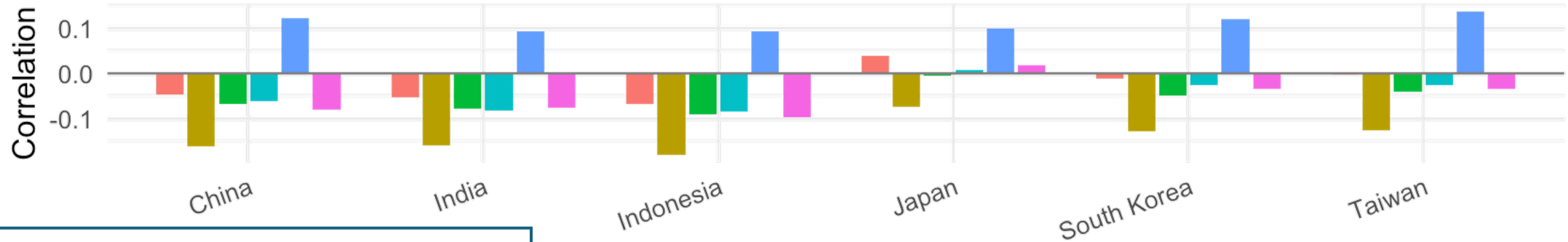
- The annual observations don't reflect the volatility in the market and ignore the current (May 2026) much higher price.
- Optimistic IMF scenarios undermine more long-term impacts that could maintain high prices.

* World Bank data. Fertilizer Index calculated by the author as the simple average of indexed prices for DAP, Phosphate Rock, Potassium Chloride, TSP, and Urea (E. Europe), with 2025 = 100



Commodity Exposure

Relationship Between Growth and Input Prices(2000-2025)



Context:

- The countries are dependent on the strait for their imports of Crude, but they are experts in refining. China is expected to lower output by 60% this year, South Korea, Japan, Taiwan and Indonesia by 30% and India by 15% (AGI). This will impact negatively the petrochemicals industry but also the downstream uses, such as transportation.
- Taiwan imports one third of its LNG from the strait (20% of its energy supply like South Korea and Japan, 60% for India, and 30% for China (but produces a lot at home). Meanwhile Indonesia is one of biggest exporters.
- Similarly, fertilizers are crucial in the chipmaking industry, which could hurt Taiwan.
- Existing speculation about the crisis benefiting China as the leader in Renewables.



General Commodity Theory:

- Even countries with their own sources of energy are impacted because prices follow global markets
- A commodity price surge can also benefit a country that sells it by strengthening its currency and thus giving it more purchasing power.
- Price increases in Commodities could also reflect a booming market with a lot of demand (and not sufficient supply) meaning that economies are doing well.

Analysis of the results:

- Mostly very low negative correlations, except for LNG's positive coefficient (albeit also low), which is explained by its massive price surge after Covid due to the recovery of activity.
- The low correlation makes it difficult to capture its effect in regression, which is why to determine the true relationship other explanatory variables of GDP Growth must be controlled for



Setup of the Regression: Principle Component Analysis

```
> summary(commodity_pca)
```

Importance of components:

	PC1	PC2	PC3
Standard deviation	1.4715	0.6848	0.6047
Proportion of Variance	0.7218	0.1563	0.1219
Cumulative Proportion	0.7218	0.8781	1.0000

```
> commodity_pca$rotation
```

	PC1	PC2	PC3
Brent Crude USD/b	0.5936064	-0.04081363	0.8037199
Food Index	0.5707087	-0.68278031	-0.4561827
Metal Index	0.5673826	0.72948289	-0.3820101

Commodity PCA Index

Constructed from Brent, Food, and Metal price growth



Commodity prices cannot be included separately in the regression analysis because it would then be affected by multicollinearity, so an index is constructed for them



Results I: Country-specific Regressions

Table 1: Commodity Exposure Regressions

	China	India	Indonesia	Japan	South Korea	Taiwan
Constant	7.847*** (0.501)	7.209*** (0.797)	7.398*** (0.374)	1.195*** (0.276)	6.228*** (1.148)	4.474*** (0.755)
Commodity PCA	0.918* (0.329)	0.070 (0.255)	0.497* (0.229)	0.309 (0.208)	1.202* (0.454)	0.978* (0.370)
Inflation	0.185 (0.118)	-0.082 (0.119)	-0.314*** (0.027)	-0.155 (0.214)	-0.716+ (0.357)	-0.096 (0.425)
Asian Crisis	2.029 (2.141)	-2.514 (1.730)	-0.157 (1.542)	0.791 (1.327)	4.635 (3.064)	2.804 (2.453)
GFC	4.121+ (2.268)	1.709 (1.913)	0.113 (1.641)	-6.530*** (1.522)	-0.295 (3.168)	-3.620 (2.706)
COVID Shock	-5.044* (2.131)	-12.409*** (1.721)	-8.324*** (1.547)	-5.170*** (1.329)	-5.331+ (3.050)	-0.092 (2.519)
COVID Recovery	-2.747 (2.440)	2.686 (1.928)	-4.986** (1.715)	1.225 (1.471)	-4.133 (3.360)	-1.066 (2.729)
Num.Obs.	31	31	31	31	31	31
R2	0.487	0.719	0.874	0.694	0.364	0.386

Always significant because of the properties of the data

Positive sign, against the common intuition, that varies in significance and weight across countries

Explanatory variable to control for other factors, such as interest rates, currency strength, social context

Time controls

Only 31 observations makes it difficult to obtain robust results

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

To obtain better results, in a country-by-country case, it is not enough to include all the variables: the different impacts that global shocks had must be accounted for by each regression



Results II: Panel Data

Accounts for country-specific factors that don't vary across time, like culture

	Country FE	Country Interactions	Advanced vs Emerging
Commodity PCA	0.711*** (0.197)	0.884*** (0.134)	0.549* (0.253)
PCA × India		-0.529*** (0.053)	
PCA × Indonesia		-0.479*** (0.107)	
PCA × Japan		-0.203*** (0.056)	
PCA × South Korea		0.003 (0.027)	
PCA × Taiwan		0.169*** (0.021)	
PCA × Advanced			0.324+ (0.194)
Inflation	-0.239** (0.073)	-0.248*** (0.073)	-0.241** (0.076)
Asian Crisis	1.261 (0.929)	1.261 (0.946)	1.261 (0.933)
GFC	-0.763 (1.673)	-0.775 (1.704)	-0.766 (1.680)
COVID Shock	-5.800*** (1.716)	-5.819** (1.739)	-5.804*** (1.722)
COVID Recovery	-1.695 (1.344)	-1.710 (1.368)	-1.698 (1.349)
Num.Obs.	186	186	186
R2	0.633	0.643	0.637

Again a positive sign, but significant

Comparison with China as the reference, seeming to indicate that the country is more resilient to commodity price increases

Would be an important result if it were significant and the impact of commodity price hikes was drawing down growth

Control variable does a better job when pooling countries

Important missing variables that come with their own limitations:

- Lag GDP growth, $Y(t-1)$: autocorrelation
- Structural components such as Debt/GDP, either accounted for in the fixed effects, or do create a conflict between short-term (commodity prices) and long-term

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001



Limitations and Opportunities for Further Research

Limitations

- A broad and complex topic that would have benefited from more time, experience, and research.
 - This is especially true for the Regression Analysis that surely included omitted variable bias from so many factors playing out at the same time.
- GDP Growth and even GDP per capita are not everything: it's also important to consider welfare and the climate.

Opportunities for further Research

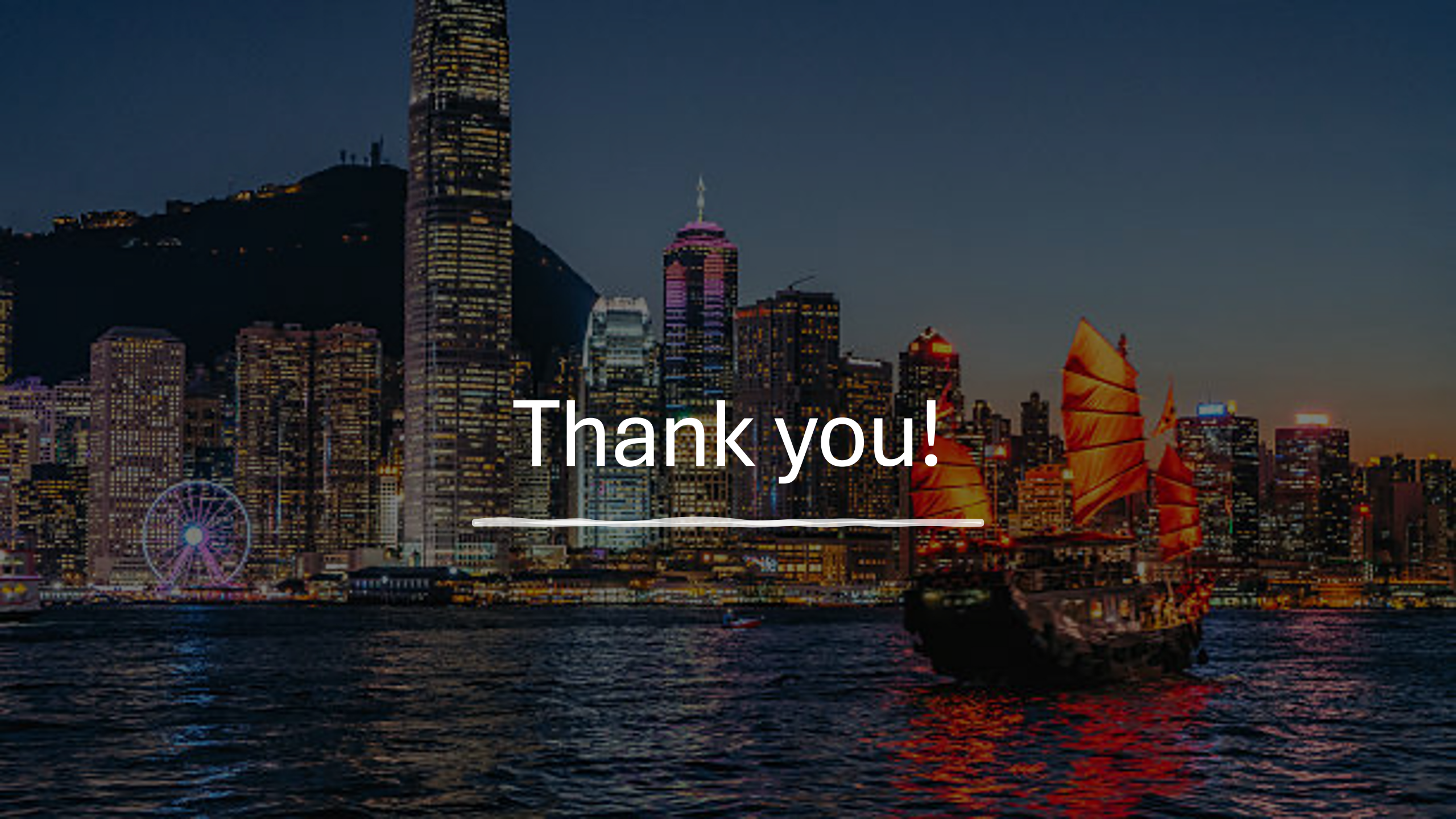
- To respond to that Regression issue, it would be important to include more indicators coming from more databases, such as Bloomberg, Stock Indices, and National Statistics Institute.
- Extend the analysis to more countries in Asia to learn more about them and even world regions to compare the expected impact of the crisis in the future.



References

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- The Strait of Hormuz crisis: Impact on supply chain and the financial market, Asia Global Institute Geoeconomics Update, Heiwai Tang, Guanzheng Sun, March 27, 2026
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- Leveraging Growth Regressions for Country Analysis, World Bank, Policy Research Working Paper 1075, Konstantin M. Wacker Robert C.M. Beyer Lars Christian Moller
- Economic Integration Report 2026: Leveraging Regional Cooperation and Integration to Navigate Global Uncertainties, Asian Development Bank, 23 February 2026



A night-time photograph of the Hong Kong skyline. The image shows a dense cluster of skyscrapers, including the prominent, illuminated Bank of China Tower on the left. The city lights reflect on the dark water in the foreground. A traditional Chinese junk boat with large, glowing red sails is positioned on the right side of the water. A Ferris wheel is visible on the far left. The sky is a deep twilight blue.

Thank you!
